

RQ Answers

omitted

2024-04-05

Data overview

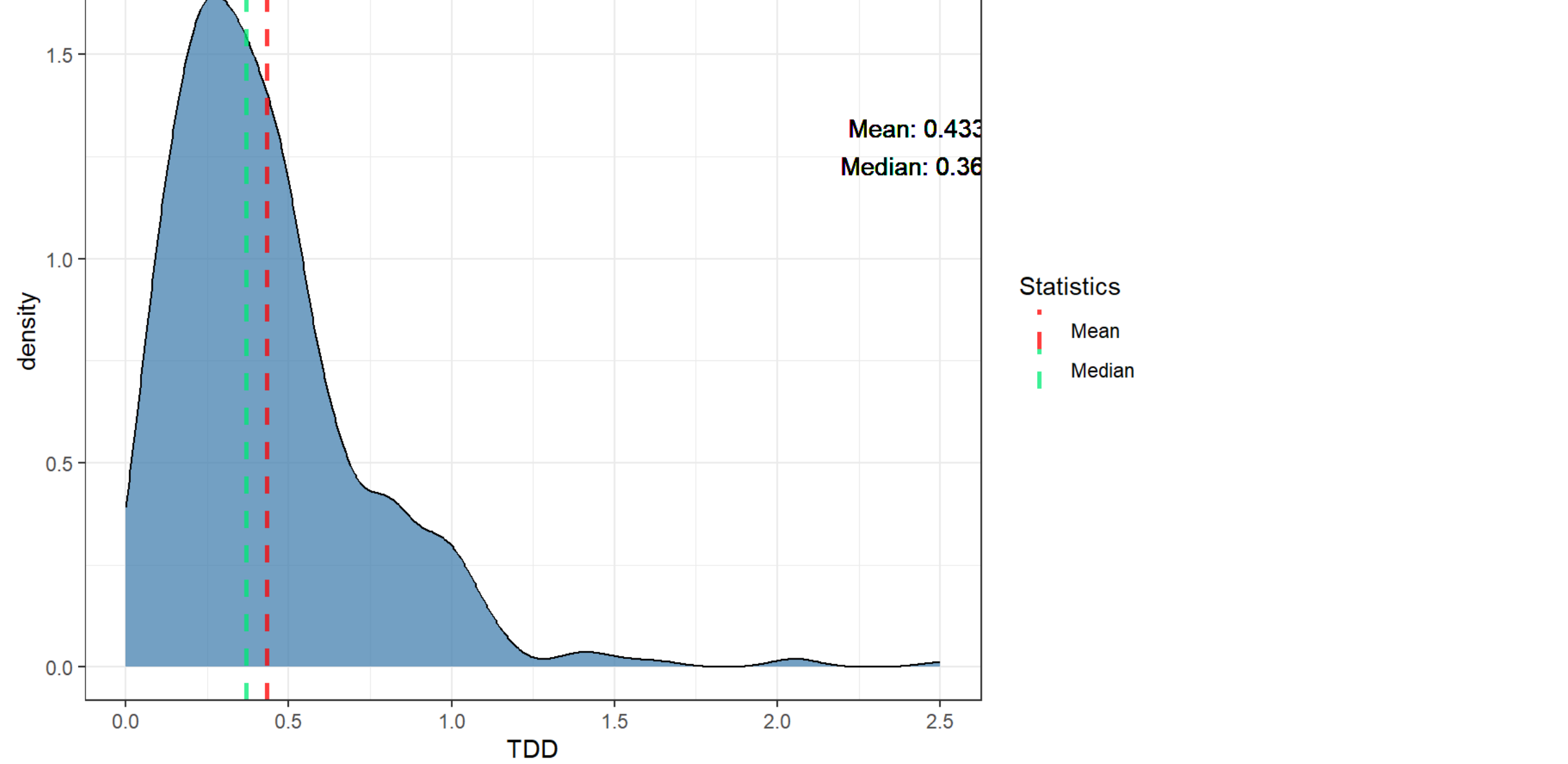
glimpse(all_data)				
## Rows: 27,414				
## Columns: 11				
## \$ repo	<chr>	"accumulo", "accumulo", "accumulo", "accumulo", "a...		
## \$ pr_number	<int>	1302, 1302, 1302, 1302, 1302, 1302, 1302, 13...		
## \$ rule	<chr>	"java:S5803", "java:S108", "java:S108", "java:S119...		
## \$ severity	<chr>	"CRITICAL", "MAJOR", "MAJOR", "CRITICAL", "CRITICA...		
## \$ file	<chr>	"server/tserver/src/test/java/org/apache/accumulo/...		
## \$ type	<chr>	"CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...		
## \$ status	<chr>	"OPEN", "CLOSED", "CLOSED", "OPEN", "OPEN", "OPEN"...		
## \$ debt	<int>	10, 5, 5, 10, 10, 10, 10, 20, 20, 10, 5, 5, 20...		
## \$ ncloc_affected_file	<int>	96, 96, 96, 96, 96, 121, 121, 249, 249, 249, ...		
## \$ complexity	<int>	7, 7, 7, 7, 6, 6, 6, 37, 37, 37, 37, 37, 37...		
## \$ origin	<chr>	"NEW", "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTI...		

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 549 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 549 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

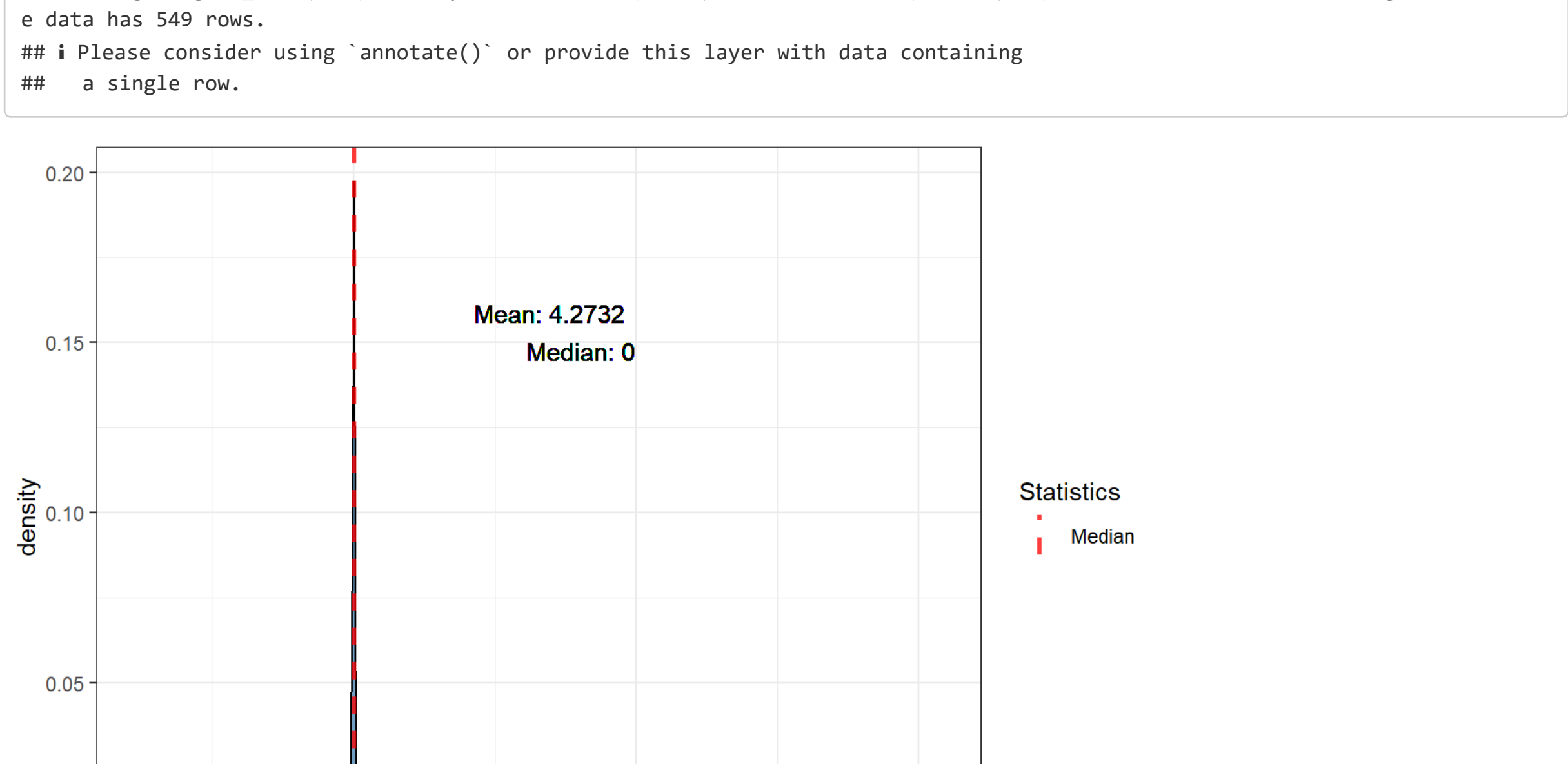


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 549 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 549 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	
## 1 accumulo	22.0	50.6	27.3	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3			
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>	<dbl>
## 1	22.0	50.6	27.3

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3			
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 accumulo	62.6	37.4	

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	62.6	37.4

Percentages for new TDV by repo

## # A tibble: 1 × 3			
## # Groups: repo [1]			
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 accumulo	83.0	17.0	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	83.0	17.0

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank accumulo		
## <int> <chr>		
## 1	1 java:S1192	- 240
## 2	2 java:S112	- 131
## 3	3 java:S1135	- 73
## 4	4 java:S100	- 52
## 5	5 java:S2095	- 39
## 6	6 java:S1155	- 35
## 7	7 java:S1659	- 28
## 8	8 java:S2259	- 25
## 9	9 java:S1854	- 24
## 10	10 java:S3776	- 22

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1192	122	
## 2 java:S112	123	
## 3 java:S1135	124	
## 4 java:S100	125	
## 5 java:S2095	126	
## 6 java:S1155	127	
## 7 java:S1659	128	
## 8 java:S2259	129	
## 9 java:S1854	130	
## 10 java:S3776	131	

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank accumulo		
## <int> <chr>		
## 1	1 java:S1192	- 7140
## 2	2 java:S112	- 3673
## 3	3 java:S1186	- 1098
## 4	4 java:S1659	- 821
## 5	5 java:S2095	- 821
## 6	6 java:S106	- 793
## 7	7 java:S100	- 666
## 8	8 java:S3252	- 648
## 9	9 java:S2184	- 600
## 10	10 java:S108	- 561

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1192	122	
## 2 java:S112	123	
## 3 java:S1186	124	
## 4 java:S1659	125	
## 5 java:S2095	126	
## 6 java:S106	127	
## 7 java:S100	128	
## 8 java:S3252	129	
## 9 java:S2184	130	
## 10 java:S108	131	

Outliers

Higher TDD

## # A tibble: 549 × 5				
## repo	pr_number	total_debt	total_ncloc	tdd
## <chr>	<int>	<chr>	<int>	<dbl>
## 1 accumulo	1768	470	188	2.5
## 2 accumulo	4103	302	145	2.08
## 3 accumulo	1424	261	129	2.02
## 4 accumulo	1616	275	167	1.65
## 5 accumulo	2752	105	68	1.54
## 6 accumulo	1735	399	273	1.46
## 7 accumulo	1498	188	134	1.40
## 8 accumulo	1304	244	174	1.40
## 9 accumulo	4007	206	155	1.33
## 10 accumulo	3756	335	285	1.18
## # i 539 more rows				

Higher TDV

## # A tibble: 549 × 3			
## pr_number	repo	tdv	
## <int> <chr>	<int>	<dbl>	
## 1	2503 accumulo	2078	
## 2	1891 accumulo	643	
## 3	1605 accumulo	574	
## 4	2335 accumulo	557	
## 5	2289 accumulo	345	
## 6	3562 accumulo	307	
## 7	2087 accumulo	293	
## 8	3639 accumulo	164	
## 9	4279 accumulo	163	
## 10	3912 accumulo	127	
## # i 539 more rows			

Lower TDV

## # A tibble: 549 × 3			
## pr_number	repo	tdv	
## <int> <chr>	<int>	<dbl>	
## 1	3604 accumulo	-768	
## 2	1565 accumulo	-704	
## 3	3117 accumulo	-571	
## 4	3402 accumulo	-360	
## 5	2435 accumulo	-350	
## 6	3344 accumulo	-190	
## 7	1304 accumulo	-174	
## 8	3229 accumulo	-173	
## 9	1424 accumulo	-170	
## 10	1714 accumulo	-132	
## # i 539 more rows			

Extra

Number of PRs with pre-existing TD

## n	
## 1 522	

Number of java:S1192 issues that affect test code

## n	
## 1 7300	

Number of java:S1192 issues that affect not test code

## n	
## 1 80	

Summary NCLOC

## ncloc	
## Min. : 16	
## 1st Qu.: 180	
## Median : 405	
## Mean : 1147	
## 3rd Qu.: 936	
## Max. : 61510	